DO NOW — RECALL

Thinking abstractly: models and reality

Name: _____ Class: _____ Date: _____

1. What is unit testing?

2. What is a precondition?

3. What do vertices and edges represent in a graph?

4. Why is an unweighted route graph not enough to identify the fastest journey?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Testing a small module or function independently.

2. Model answer

An input requirement that must hold before execution.

3. Model answer

Objects/locations and connections.

4. Model answer

It lacks travel-time weights.

Vocabulary — know these terms

abstraction — Retain relevant characteristics, ignore irrelevant ones.

model — Simplified representation serving a purpose.

attribute — Property stored about an entity.

assumption — Condition accepted in constructing the model.

limitation — A question the simplified model cannot answer.

NEW LEARNING

Read, annotate and explain

Choose a purpose

Decide whether a particular book may be loaned.

Keep essential facts

ISBN identifies the book, and Available indicates whether it can be borrowed.

Evaluate omissions

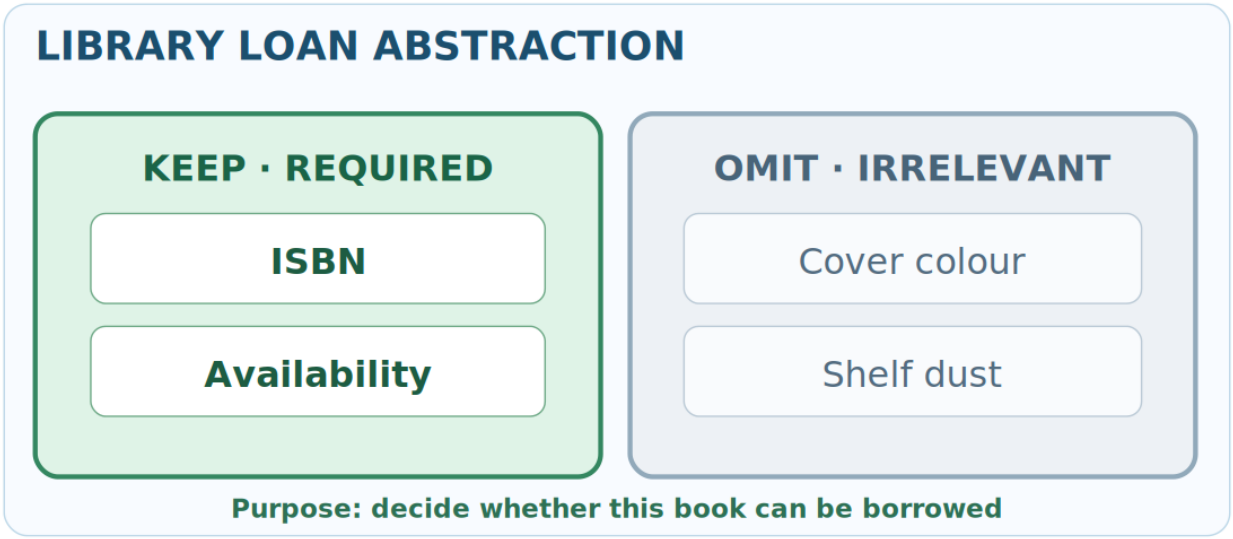
The model cannot answer questions about book weight or appearance because those attributes are omitted.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Library loan abstraction: keep ISBN and availability; omit cover colour and dust

New Learning

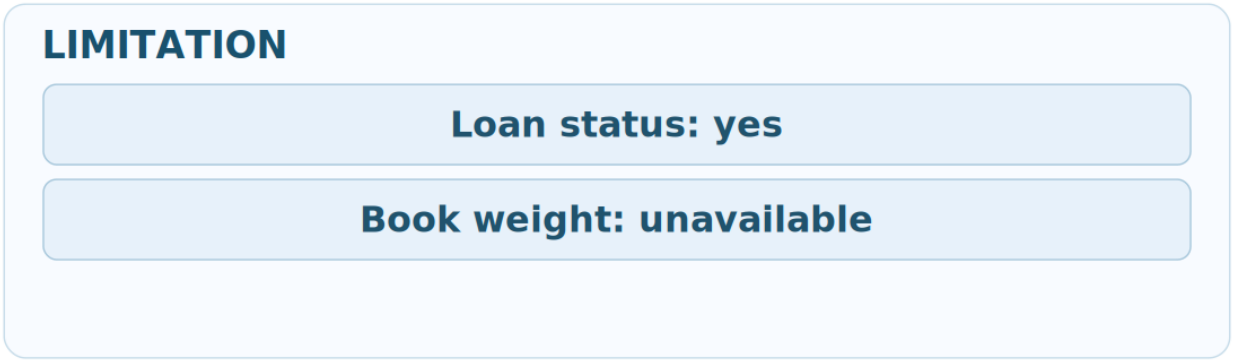


PURPOSE; Find title; Check availability; Record loan

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



LIMITATION; Loan status: yes; Book weight: unavailable

I DO — WORKED EXAMPLE

Follow the worked method

Model a bus network for finding the route with the fewest stops.

Predict first

Model each bus stop as a vertex and each permitted connection as an edge.

Model and check

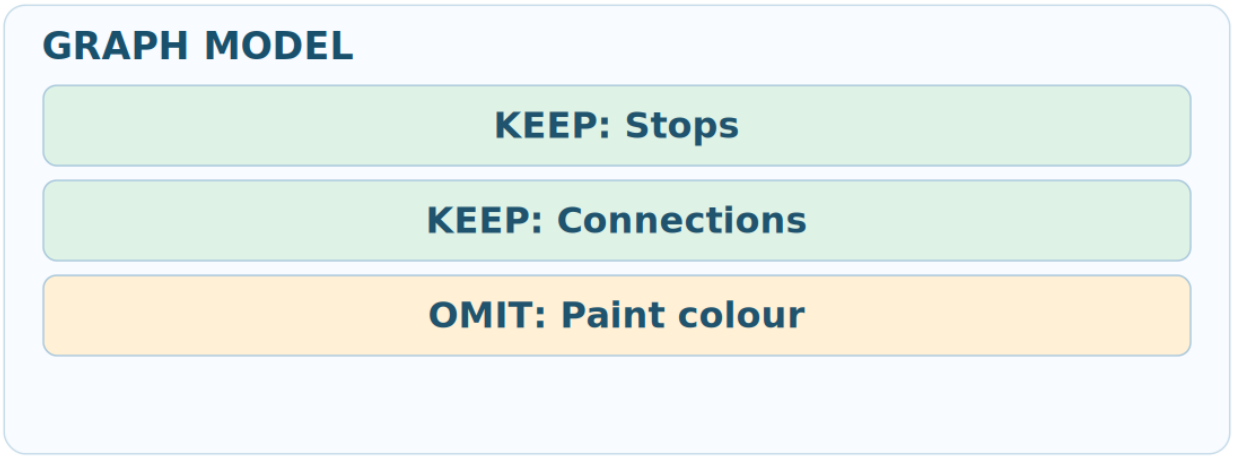
Vehicle colour is irrelevant. To find fastest rather than fewest-stop routes, add journey-time weights.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Model a bus network for finding the route with the fewest stops.

Model each bus stop as a vertex and each permitted connection as an edge.



GRAPH MODEL; KEEP: Stops; KEEP: Connections; OMIT: Paint colour

Model a bus network for finding the route with the fewest stops.

Model each bus stop as a vertex and each permitted connection as an edge.



ROUTE; Stop A; Stop B; Stop D

WE DO — GUIDED PRACTICE

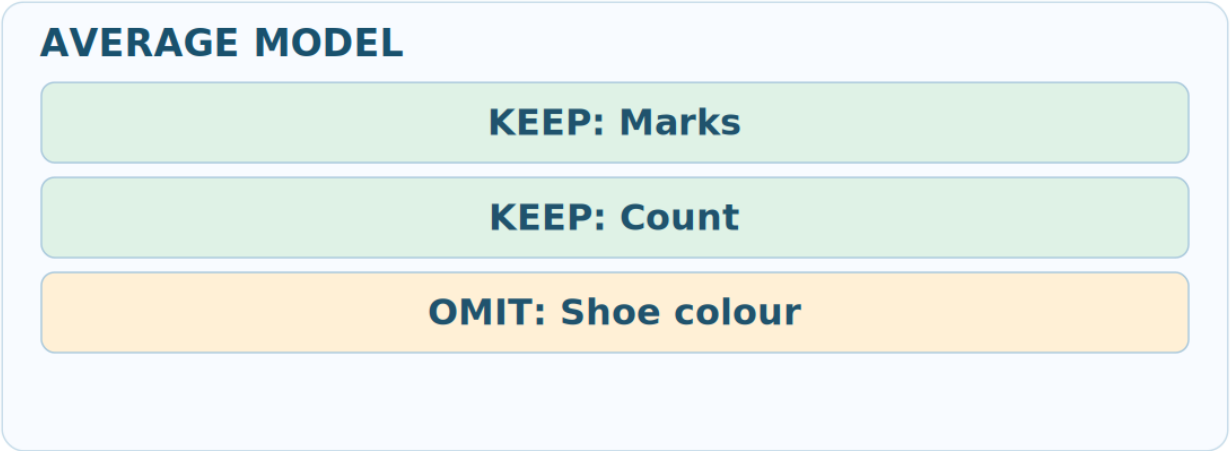
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Model the average of a class's valid test marks.

Only valid marks and the number of entries are needed to compute the mean.



AVERAGE MODEL; KEEP: Marks; KEEP: Count; OMIT: Shoe colour

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first Only valid marks and the number of entries are needed to compute the mean.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Identify two relevant and two irrelevant attributes for a stock checker.

Question 2

Give a simple abstraction for detecting school room timetable clashes: three needed fields and two omitted details.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Why should a model not contain every real-world detail? Give an example where a previously omitted detail becomes relevant.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A good abstraction must record every real-world property.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

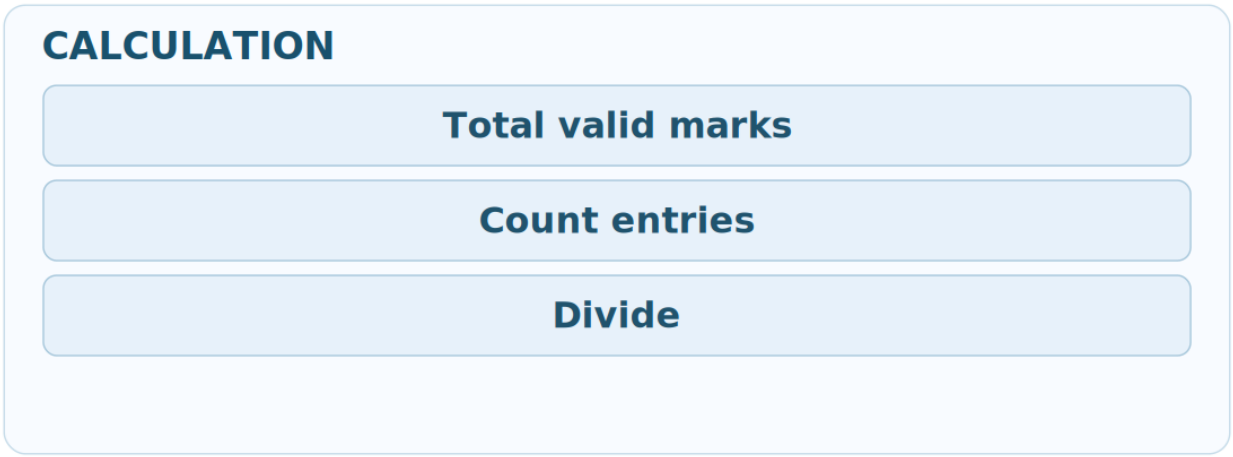
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Model the average of a class's valid test marks.

Only valid marks and the number of entries are needed to compute the mean.



CALCULATION; Total valid marks; Count entries; Divide

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model and check

Define how to handle missing results. Other pupil attributes need not be stored for this calculation.

IMPROVE: Explain one correction to your own reasoning.